

The Earth

OUR PLANET IS THE THIRD FROM THE SUN AND ONE OF FOUR IN THE SOLAR SYSTEM MADE MAINLY OF ROCK AND METAL.

Earth is the only planet where liquid water is known to exist. It is also the only place in the Universe known to support life.

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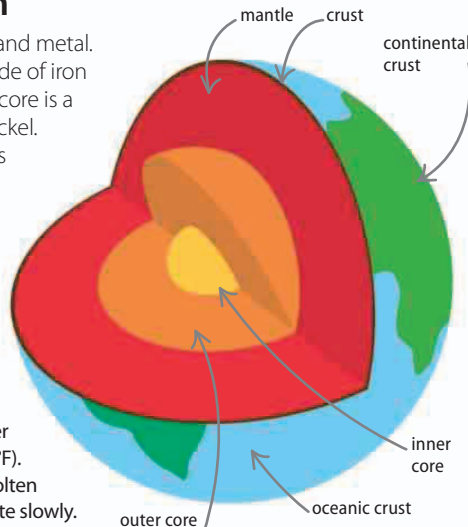
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Inside the Earth

Earth is a mixture of rock and metal. The solid inner core is made of iron and nickel, and the outer core is a mix of molten iron and nickel. The surrounding mantle is a thick layer of solid and semimolten rock. A thin, rocky outer shell consists of thick continental crust (land) and thinner oceanic crust (seafloor).

▷ Inner heat

Inside, the Earth is very hot. The temperature at the inner core reaches $4,700^{\circ}\text{C}$ ($8,500^{\circ}\text{F}$). The heat causes the semimolten rock in the mantle to circulate slowly.



chain of extinct volcanoes left behind as plates move away from "hot spot"

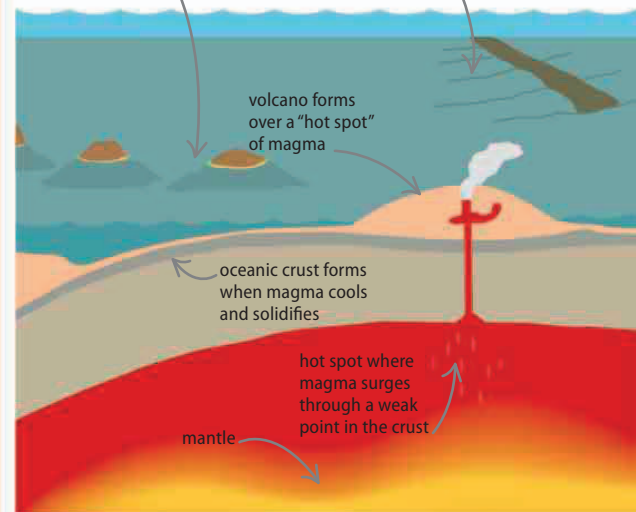
spreads out where two tectonic plates move apart

volcano forms over a "hot spot" of magma

oceanic crust forms when magma cools and solidifies

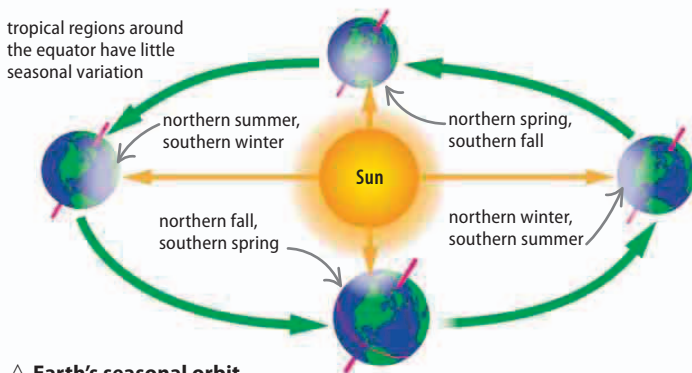
hot spot where magma surges through a weak point in the crust

mantle



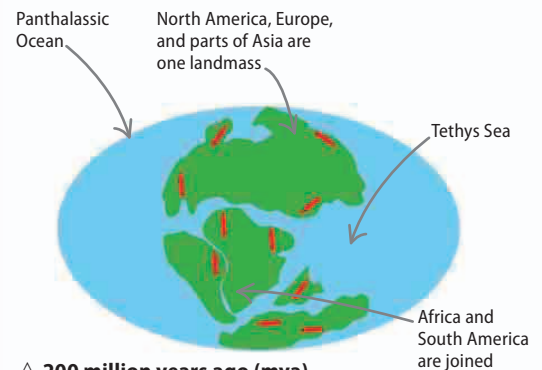
The seasons

Earth experiences seasons because it rotates on a tilted axis as it travels around the Sun, an orbit that takes one year. As different areas of the planet face toward or away from the Sun, the length of the day and temperature change, which affect plant growth and animal behavior.



△ Earth's seasonal orbit

When the North Pole turns to face the Sun it is summer in the Northern Hemisphere and winter in the south. Six months later, when the South Pole tilts toward the Sun, it is summer in the south and winter in the north.



△ 200 million years ago (mya)

From a single landmass called Pangaea, the continents begin breaking apart. The northern continents become separated from those in the south by the Tethys Sea.